



Engineered exosome-mediated delivery of MicroRNA let-7i-5p targets toll-like receptor 4 to modulate inflammation and alleviate acute liver injury in septic mice

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ABSTRACT

Engineered exosomes have emerged as a promising platform for the targeted delivery of therapeutic nucleic acids. The microRNA let-7i-5p is recognized for its anti-inflammatory properties, but its molecular targets remain incompletely defined. To elucidate its mechanism of action and facilitate its efficient delivery, we generated let-7i-5p-enriched engineered exosomes (EEXO) by overexpressing let-7i-5p in genetically modified RAW264.7 cells. Droplet digital polymerase chain reaction confirmed that EEXO treatment significantly increased the intracellular level of let-7i-5p (3-fold increase; $p < 0.001$). Bioinformatic analyses performed using four microRNA target prediction databases identified toll-like receptor 4 (TLR4) as a potential target. This finding was validated by a luciferase activity assay, which revealed direct binding of let-7i-5p to the 3' untranslated region of TLR4 mRNA, resulting in its suppression ($p < 0.001$). In lipopolysaccharide (LPS)-stimulated macrophages, EEXO-delivered let-7i-5p downregulated TLR4 and its downstream signaling components, such as myeloid differentiation primary response 88 (MyD88), tumor necrosis factor receptor-associated factor 6 (TRAF6), and nuclear factor- κ B (NF- κ B), thereby reducing the levels of proinflammatory cytokines such as tumor necrosis factor- α and interleukin-6 (all $p < 0.05$). In vivo, systemic EEXO administration attenuated LPS-induced liver injury in adult male mice, as indicated by reduced histopathological damage, liver enzyme release, and tissue edema (all $p < 0.05$). Furthermore, EEXOs suppressed the TLR4/MyD88/TRAF6/NF- κ B pathway and proinflammatory cytokine production in liver tissues. In conclusion, engineered exosome-delivered let-7i-5p targets TLR4 and modulates MyD88/TRAF6/NF- κ B signaling to alleviate inflammation and protect against LPS-induced liver injury.

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1. Introduction

Let-7i-5p is a small, evolutionarily conserved microRNA belonging to the let-7 family, which plays a central role in post-transcriptional gene regulation [1]. MicroRNAs such as let-7i-5p bind to complementary sequences within the 3' untranslated regions (3'UTRs) of target messenger RNAs (mRNAs), leading to mRNA degradation or translational repression [2]. This finely tuned regulation maintains cellular homeostasis and balanced gene expression [3,4]. The strong sequence conservation of let-7i-5p between humans and mice underscores its fundamental biological importance [5].

Functionally, let-7i-5p participates in diverse biological processes, including cell proliferation, differentiation, apoptosis, immune modulation, and inflammation [1,6]. Its expression is tightly controlled during development and in adult tissues, reflecting its broad physiological significance [6]. Dysregulation of let-7i-5p has been linked to various diseases, particularly cancer, where it may act as either a tumor suppressor or an oncogene depending on cellular context [7]. Aberrant expression contributes to cell-cycle imbalance, epithelial-mesenchymal transition, metastasis, and chemoresistance [6]. Beyond malignancy, altered let-7i-5p levels have been reported in cardiovascular, neurodegenerative, and immune-mediated disorders [8–11], highlighting its wide-ranging pathophysiological relevance.

Recent evidence has underscored the anti-inflammatory and cytoprotective properties of let-7i-5p [10–13]. Overexpression of this microRNA alleviates inflammation, mitochondrial dysfunction, and apoptosis in hypoxia-treated cardiomyocytes [13]. Our previous studies identified let-7i-5p as the most abundant microRNA in exosomes derived from human placental mesenchymal stem cells (MSCs) and demonstrated its critical contribution to the therapeutic effects of MSC-exosomes against lipopolysaccharide (LPS)-induced organ injury in both obese and non-obese mice [10]. Moreover, engineered exosomes enriched with let-7i-5p (EEXOs) achieved therapeutic efficacy comparable to MSC-exosomes in murine sepsis models [11], underscoring the translational potential of this microRNA.

Given that stable and targeted delivery remains a major challenge for RNA-based therapeutics, engineered exosomes have emerged as an attractive vehicle for microRNA delivery. Their biocompatibility, low immunogenicity, and ability to traverse biological barriers confer clear advantages over conventional delivery systems [14–17]. Engineered exosomes can be tailored to encapsulate specific cargos—such as microRNAs or small-molecule drugs—while protecting them from enzymatic degradation and enabling precise tissue targeting [14–17]. These properties make them a next-generation platform for delivering fragile nucleic acids with enhanced stability and therapeutic specificity both *in vitro* and *in vivo* [14–17].

To elucidate the molecular mechanisms underlying the anti-inflammatory effects of let-7i-5p, this study employed an integrated approach combining bioinformatic prediction, *in vitro* validation, and *in vivo* functional assessment. Candidate targets were identified using multiple microRNA prediction databases and validated through luciferase reporter assays and gene expression analyses. The *in vivo* relevance of these findings was examined using a murine model of LPS-induced liver injury. Engineered exosomes were generated by plasmid-mediated overexpression of let-7i-5p in RAW264.7 macrophages to ensure efficient delivery. To our knowledge, this is the first study to systematically delineate the molecular basis of let-7i-5p-mediated inflammation control, highlighting its potential as a therapeutic modulator of inflammatory disorders.

2. Methods

2.1. Generation and characterization of EEXOs

2.1.1. Generation and isolation of EEXOs

EEXOs were produced by genetically modifying RAW264.7 cells

through plasmid-mediated transfection to overexpress let-7i-5p, following established protocols [11]. No exosomal protein tag-based targeting system was used; instead, let-7i-5p was enriched in exosomes through the cells' intrinsic RNA packaging mechanisms, which efficiently incorporated overexpressed microRNAs into secreted vesicles.

RAW264.7 cells (Bioresource Collection and Research Center, Hsinchu, Taiwan) were cultured in Dulbecco's Modified Eagle Medium (Sigma-Aldrich, St. Louis, MO, USA) supplemented with 10 % fetal bovine serum (Nihonri Biosciences, Tokyo, Japan) plus penicillin (100 U/mL) and streptomycin (100 µg/mL; both from Gibco, El Paso, TX, USA). The cells were maintained at 37 °C under 5 % CO₂ in a humidified incubator.

To initiate transfection, confluent RAW264.7 cells were transferred to serum-free medium (Thermo Fisher, Waltham, MA, USA) and incubated for 30 min. Then, the cells were transfected with 2 µg of a let-7i-5p-encoding plasmid (5'-UGAGGUAGUAGUUUGUGCUGUU-3'; pCMV-let-7i-5p, SC400011; OriGene Technologies, Rockville, MD, USA). After a 5-min incubation, 205 µL of serum-free medium containing 5 µL of Lipofectamine 3000 (Thermo Fisher) was added. Cells stably overexpressing let-7i-5p were subsequently selected using neomycin. EEXOs were isolated from the culture medium of stably transfected cells by using established protocols [11]. In brief, the medium was collected and centrifuged at 300 ×g for 10 min at 4 °C (Allegra X-15R; Beckman Coulter, Brea, CA, USA). The supernatant was passed through a 0.22-µm filter (Millipore, Burlington, MA, USA) to remove debris and then subjected to two rounds of ultracentrifugation at 100,000 ×g for 90 min at 4 °C (Optima L-80XP Ultracentrifuge; Beckman Coulter) by using a Type 50.2 Ti rotor (k-factor: 157.7). The resulting EEXO pellets were resuspended in 500 µL of phosphate-buffered saline (PBS; Thermo Fisher) and stored at −80 °C for future use.

2.1.2. Characterization of EEXOs

2.1.2.1. Morphology, size distribution, and exosome marker analysis. EEXOs were characterized using established protocols [11]. Morphological features of EEXOs were explored through transmission electron microscopy (TEM). For TEM, 3 µL of EEXO suspension was fixed with 50 µL of 2 % paraformaldehyde (Sigma-Aldrich) and placed onto Formvar-carbon-coated electron microscopy grids. The samples were examined using a JEM 1400 series transmission electron microscope (JEOL, Tokyo, Japan).

Size distribution and particle concentration were determined through nanoparticle tracking analysis by using the NanoSight NS300 instrument (Malvern Panalytical, Malvern, UK). Measurements were performed following the manufacturer's protocols to ensure accuracy and reproducibility.

Expression of the exosome markers ALIX, CD9, and CD63 was evaluated through immunoblotting [18]. In brief, 20 µg of protein from EEXO samples was separated on 12 % sodium dodecyl sulfate polyacrylamide gels (Sigma-Aldrich), transferred onto nitrocellulose membranes (Bio-Rad Laboratories, Hercules, CA, USA), and incubated overnight with primary antibodies against ALIX (ab235377; Abcam, Cambridge, UK), CD9 (IR300–981; iReal Technology, Hsinchu, Taiwan), and CD63 (IR301–983; iReal Technology). Images were acquired using a multispectral imaging system (UVP BioSpectrum 815; Analytik Jena, Jena, Germany).

2.1.2.2. Measurement of let-7i-5p concentration in EEXOs. The concentration of let-7i-5p in EEXOs was quantified through droplet digital polymerase chain reaction (ddPCR), following established methods [11]. Total RNA was extracted from EEXOs by using the miR-Neasy Serum/Plasma Kit (Qiagen, Hilden, Germany), following the manufacturer's protocol. Reverse transcription was performed using the TaqMan® MicroRNA Assay (Thermo Fisher). The copy number of let-7i-5p was measured using the QX200 ddPCR system (Bio-Rad). Reverse

transcription products, primers, master mix, and mineral oil were loaded into a droplet generator to produce thousands of droplets, followed by polymerase chain reaction (PCR) with the TaqMan® MicroRNA Assay (Thermo Fisher). After amplification, the droplets were aspirated and analyzed using the Droplet Reader (Bio-Rad). Data were analyzed using QuantaSoft (Bio-Rad). Blank exosomes derived from unmodified RAW264.7 cells (BEXOs) were used as controls, and let-7i-5p concentrations were compared between BEXOs and EEXOs.

2.2. Bioinformatic analysis to predict the potential targets of let-7i-5p

Bioinformatic analyses were performed to predict the mRNA targets of let-7i-5p by using four established algorithms: DIANA-microT 2023, miRPATHDP, miRTarBase, and MiRanda. These algorithms identified candidate genes by analyzing complementary binding sites and regulatory interactions. Raw data from each algorithm were refined and consolidated using bioinformatics tools available at MolBioTools (<http://s://molbiotools.com>). To identify overlapping targets, the results obtained using the four algorithms were compared, and shared targets were visualized in a Venn diagram. After the initial screening, focused selection was performed to prioritize genes involved in inflammatory pathways. This step unveiled key let-7i-5p targets whose regulation significantly modulated immune responses and inflammation. Binding sites of let-7i-5p within the 3'UTR sequences of predicted targets were also identified, accounting for both Watson-Crick base pairing (|) and G•U wobble base pairing (:) [19]. The analysis indicated oll-like

receptor 4 (TLR4) as the key target of let-7i-5p (Fig. 1), given the central role of this protein in the recognition of pathogens, activation of inflammatory pathways, and regulation of immune responses [20–22].

2.3. In vitro experiments

2.3.1. Determination of the optimal doses of EEXOs and LPS for in vitro experiments

In vitro experiments were conducted using RAW264.7 cells with or without LPS stimulation [23]. To determine the optimal doses of EEXOs and LPS, RAW264.7 cells were seeded into a 96-well plate and treated with PBS (Sigma-Aldrich), EEXOs (1×10^8 or 1×10^9 particles/mL), or LPS (50, 100, or 500 ng/mL; *Escherichia coli* O127:B8; Sigma-Aldrich). After 24 h of incubation, cell viability was measured using the cell counting kit-8 assay (Sigma-Aldrich) at 450 nm with a microplate reader (Molecular Devices, San Jose, CA, USA) following the manufacturer's instructions. The results revealed that at neither dose— 1×10^8 or 1×10^9 particles/mL, EEXOs affected cell viability (Fig. 3A). Among the LPS concentrations, 100 ng/mL was the highest dose that did not significantly reduce viability (Fig. 3A). Therefore, this concentration was selected for subsequent in vitro experiments.

The optimal dose of EEXOs for in vitro experiments was determined using the LPS-stimulated RAW264.7 model. RAW264.7 cells were treated with PBS, LPS (100 ng/mL), or LPS plus EEXOs (1×10^8 or 1×10^9 particles/mL). After 24 h, culture supernatants were collected, and interleukin-6 (IL-6) levels were measured using an enzyme-linked

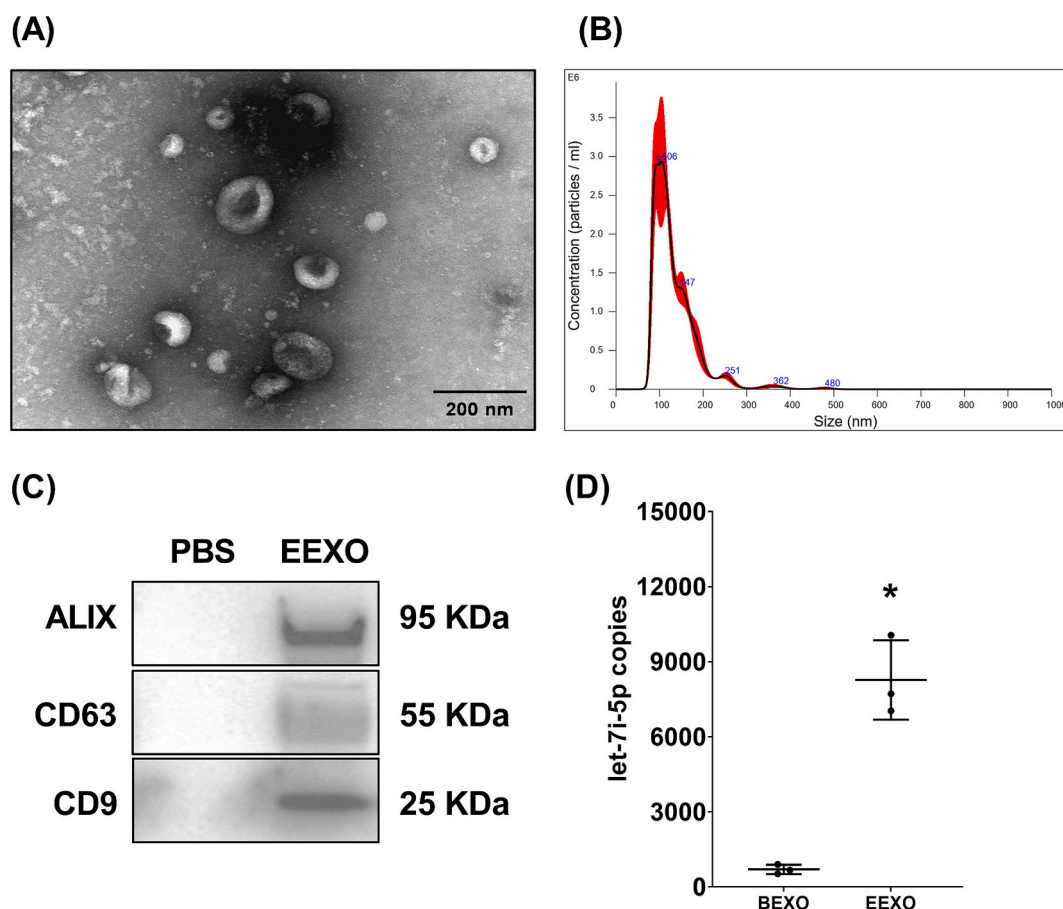


Fig. 1. Characteristics of engineered exosomes enriched with microRNA let-7i-5p (EEXO), isolated from genetically modified RAW264.7 cells overexpressing let-7i-5p. (A) Representative transmission electron microscopy images of isolated EEXO. (B) Particle size distribution of isolated EEXO, measured using nanoparticle tracking analysis. (C) Representative immunoblot images showing the presence of markers ALIX, CD63, and CD9 in isolated EEXO. (D) Concentrations of let-7i-5p in isolated EEXO and blank exosomes from unmodified RAW264.7 cells (BEXO), measured using droplet digital polymerase chain reaction. Data are presented as mean $\pm$ standard deviations ($n = 3$ per group) and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, the EEXO group versus the BEXO group.

immunosorbent assay (ELIZA) kit (IL-6 ELISA kit, ab222503; Abcam, Cambridge, UK), following the manufacturer's protocol. Of the two EEXO doses tested, 1×10^9 particles/mL significantly attenuated LPS-induced upregulation of IL-6 expression (Fig. 3B). Thus, this dose was selected for subsequent in vitro experiments.

The combined effect of LPS and EEXOs on cell viability was also evaluated. RAW264.7 cells seeded in 96-well plates were treated with LPS (100 ng/mL) or LPS (100 ng/mL) plus EEXOs (1×10^9 particles/mL) for 24 h. Cell viability was measured using the cell counting kit-8 assay, as described in the previous section.

2.3.2. In vitro biobehavior analysis of EEXOs

2.3.2.1. EEXO uptake by RAW264.7 cells. EEXO uptake by RAW264.7 cells was evaluated through confocal microscopy. EEXOs were labeled using the ExoGlow-Protein EV Labeling Kit (System Biosciences, Palo Alto, CA, USA) [24], following the manufacturer's instructions, and stored at -20°C until use. Confluent RAW264.7 cells were seeded into glass-bottom confocal dishes (SPL 200350; SPL Life Sciences, Pocheon, Korea) and cultured as described previously. The cells were stained with Hoechst 33342 (Thermo Fisher) for the nuclei [25] and PlasMem Bright dye (Dojindo Lab., Kumamoto, Japan) for plasma membranes [26], following the manufacturers' protocols. After two PBS washes to remove excess dye, the cells were incubated in Dulbecco's Modified Eagle Medium (37°C ; 5 % CO_2) for 2 h. Labeled EEXOs (1×10^9 particles/mL) were then coincubated with the cells for either 1 or 12 h before the medium was replaced. The uptake of EEXOs by RAW264.7 cells was visualized using a Stellaris 8 confocal microscope (Leica Microsystems, Wetzlar, Germany) with one channel each for Hoechst 33342, ExoGlow-Protein, and PlasMem Bright dye.

2.3.2.2. Analysis of the effects of EEXO-mediated let-7i-5p delivery. The effects of EEXO-mediated let-7i-5p delivery to RAW264.7 cells were analyzed through reverse transcription quantitative PCR (qPCR) [27]. RAW264.7 cells were treated with PBS or EEXOs (1×10^9 particles/mL) for 12 h. Total RNA was extracted using RNeasy Lysis Buffer (Qiagen, Crawley, UK) and RNeasy Spin Columns (Qiagen, Crawley, UK). Reverse transcription was performed using microRNA SLRT Primer (TopGen Biotechnology, Kaohsiung, Taiwan) and HiScript I Reverse Transcriptase (20,000 U/100 μL ; Bionovas Technology, Toronto, Canada), following the manufacturers' instructions. The reaction conditions were as follows: denaturation at 25°C for 5 min, reverse transcription at 42°C for 30 min, and enzyme inactivation at 85°C for 5 min.

Expression levels of let-7i-5p and the internal control small nuclear RNA U6 were quantified using the Let-7i-5p_hsa_TopQ miRNA Probe qPCR Assay (602001_COA_TPGN_1226Y24; TopGen Biotechnology) and the U6-snRNA_Hs-Mm-Rn_TopQ miRNA Probe qPCR Assay (602001_COA_TPGN_1226Y24; TopGen Biotechnology). Primer sequences for let-7i-5p and U6 are listed in Table 1. qPCR was performed using RealQ Plus 2 $\times$ Master Mix for Probe (Ampliqon A/S, Odense, Denmark) on the StepOnePlus Real-Time PCR System (AB Applied Biosystems, Thermo Fisher), following the manufacturers' protocol. The thermal cycling conditions were as follows: initial enzyme activation at 95°C for 15 min, followed by 40 cycles of denaturation at 95°C for 3 s and annealing at 60°C for 40 s. Each sample was analyzed in triplicate, and the threshold cycle (Cq) values were averaged. Relative expression levels were calculated using the $2^{-\Delta\Delta\text{Cq}}$ method.

2.3.3. Luciferase activity assay to explore the interaction between let-7i-5p and TLR4 mRNA

The interaction between let-7i-5p and its predicted target, TLR4 mRNA, was explored using a luciferase activity assay, following established protocols [28]. Plasmids containing the control vector (pMir-Target), the 3'UTR sequence of wild-type TLR4 mRNA (pMirTLR4), or the 3'UTR sequence of mutant TLR4 mRNA (pMirTLR4mut) were

Table 1

Primers used in reverse transcription quantitative polymerase chain reaction.

	Forward	Reverse
Let-7i-5p ^a	5'-CGCGCGTGAGGTAGTAGTTGT-3'	5'-AGTGCAGGGTCCGAGGTATT-3'
U6 ^b	5'-CTCGCTTCGGCAGCATATAT-3'	5'-ATGGAACGCTTCACGAATTTG-3'
TLR4 [30]	5'-TTC AGA GCC GTT GGT GTA TC-3'	5'-CCC ATT CCA GGT TAG GTG TTT-3'
MyD88 [32]	5'-GTT GTG TGT GTC CGA CCG T-3'	5'-GTC AGA AAC AAC CAC CAC CAT GC-3'
TRAF6 [32]	5'-CAT CTT CAG TTA CCG ACA GCT CAT-3'	5'-TGG TCG AGA ATT GTA AGG CGT AT-3'
NF- κ B [32]	5'-CCA AAG AAG GAC ACG ACA GAA TC-3'	5'-GGC AGG CTA TTG CTC ATC ACA-3'
TNF- α [33]	5'-AAT GGC CTC CCT CTC ATC AGT T-3'	5'-CCA CTT GGT GGT TTG CTA CGA-3'
IL-6 [33]	5'-GAA CAA CGA TGA TGC ACT TGC-3'	5'-TCC AGG TAG CTA TGG TAC TCC-3'
GAPDH [31]	5'-ACC ACA GTC CAT GCC ATC AC-3'	5'-TCC ACC ACC CTG TTG CTG TA-3'

Abbreviations: TLR4, Toll-like receptor 4; MyD88, myeloid differentiation primary response 88; TRAF6, tumor necrosis factor receptor-associated factor 6; NF- κ B, nuclear factor- κ B; TNF- α , tumor necrosis factor- α ; IL-6, interleukin-6; GAPDH, glyceraldehyde-3-phosphate dehydrogenase.

^a Primer sequences for let-7i-5p were obtained using the Let-7i-5p_hsa_TopQ miRNA Probe qPCR Assay (TopGen Biotechnology, Kaohsiung, Taiwan).

^b Primer sequences for U6 were obtained from the U6-snRNA_Hs-Mm-Rn_TopQ miRNA Probe qPCR Assay (TopGen Biotechnology, Kaohsiung, Taiwan).

prepared using the Geneaid Midi Plasmid Kit (Geneaid Biotech, New Taipei City, Taiwan), following the manufacturer's instructions.

The pMirTLR4 plasmid was generated by amplifying the 3'UTR region of wild-type TLR4 mRNA from RAW264.7 complementary DNA (cDNA) by using the primers mmu-TLR4_Foward (5'-CGGAATTCGGAG AAAAACTCTGGGG-3') and mmu-TLR4_Reverse (5'-CGACGCGTT-GAAGACCTCTC-3'). The purified PCR product was digested with *Eco*RI and *Mlu*I (both from Thermo Fisher) and inserted into the pMirTarget vector to generate pMirTLR4. To construct the pMirTLR4mut plasmid, a megaprimer-based PCR mutagenesis strategy [29] was adopted using pMirTLR4 as the template. The primers LUC_Foward (5'-CCAGCCAGGTTACAACCGCC-3') and mmu-TLR4 mut-Reverse (5'-CTTTTCCTAGATATCATATTCATACCC-3') were used for amplification. The amplified product was digested with *Eco*RI and *Mlu*I and ligated into the pMirTarget vector to yield pMirTLR4mut.

RAW264.7 cells were seeded into 96-well plates and transfected with pMirTarget, pMirTLR4, or pMirTLR4mut by using Lipofectamine 2000 (Thermo Fisher), following the manufacturer's protocol. After 24 h, the cells were incubated with PBS or EEXOs (1×10^9 particles/mL) and then with a luminescence reagent for firefly luciferase detection (Neolite Reporter Gene Assay; Revvity, Waltham, MA, USA). Luminescence was measured using a SpectraMax iD5 microplate reader (Molecular Devices).

2.3.4. Effects of let-7i-5p on LPS-induced TLR4-mediated signaling in vitro

2.3.4.1. Effects of let-7i-5p on TLR4 mRNA in LPS-stimulated RAW264.7 cells. RAW264.7 cells were treated with PBS, LPS (100 ng/mL), or LPS plus EEXOs (1×10^9 particles/mL), with EEXOs administered 1 h before LPS stimulation. The cells were harvested at 6, 12, and 24 h after LPS stimulation, and lysates were prepared as described previously. TLR4 mRNA expression was quantified through qPCR. Total RNA extraction, cDNA synthesis, and qPCR were performed as described previously. Primer sequences for TLR4 [30] and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) [31], the internal control, are listed in Table 1. The primers were synthesized by Integrated DNA Technologies

(Coralville, IA, USA). Relative expression of TLR4 was calculated using the $2^{-\Delta\Delta C_q}$ method.

2.3.4.2. Generation of let-7i-5p inhibitor-treated EEXOs through electroporation. To investigate the role of let-7i-5p, EEXOs treated with a specific let-7i-5p inhibitor was generated through electroporation, following established methods [11]. The inhibitor (sequence: 5'-AACAGCACAAACUACUACCUCA-3'; Biotools, New Taipei City, Taiwan) was introduced into EEXOs through electroporation. The procedure included precipitation, resuspension, transfer, electroporation (150 V/100 μ F), unincorporated microRNA removal, and ultracentrifugation. The resulting pellets of let-7i-5p inhibitor-treated EEXOs were resuspended and stored at -80°C until use.

2.3.4.3. Analysis of the effects of let-7i-5p on the downstream signaling of TLR4 in LPS-stimulated RAW264.7 cells. RAW264.7 cells were treated with PBS, EEXOs (1×10^9 particles/mL), LPS (100 ng/mL), LPS plus EEXOs, or LPS plus let-7i-5p inhibitor-treated EEXOs (1×10^9 particles/mL). After 24 h of incubation, the cells were harvested, processed, and lysed for analysis. Total RNA was extracted from the cell lysates, and cDNA synthesis and qPCR were performed as described previously. We measured the transcript levels of TLR4 and its downstream signaling components, such as myeloid differentiation primary response 88 (MyD88), tumor necrosis factor (TNF) receptor-associated factor 6 (TRAF6), and nuclear factor- κ B (NF- κ B), as well as the proinflammatory cytokines TNF- α and IL-6. Primer sequences for TLR4 [30], MyD88 [32], TRAF6 [32], NF- κ B [32], TNF- α [33], and IL-6 [33] are listed in Table 1. The primers were synthesized by Integrated DNA Technologies. GAPDH [31] served as the internal control. Relative expression levels were calculated using the $2^{-\Delta\Delta C_q}$ method.

2.4. In vivo experiments

2.4.1. Animals and animal care

All animal experiments were approved by the Institutional Animal Use and Care Committee of Taipei Medical University, Taiwan (approval number: LAC-2020-0373). For the experiments, adult male wild-type C57B/L6 mice (age: 7 to 8 weeks) were obtained from Taiwan National Laboratory Animal Center (Taipei, Taiwan). The mice were housed under a 12-h light/dark cycle. All care and handling procedures adhered to the guidelines of the US National Institutes of Health.

Sex-based differences in sepsis outcomes are well documented. Clinical evidence indicates that men have a higher risk of poor prognosis than do women [34]. Experimental models consistently demonstrated greater sepsis susceptibility in male mice than in female mice [35]. Accordingly, only male mice were included in the present study to model the higher-risk population and ensure consistency across in vivo experiments.

2.4.2. Effects of let-7i-5p on LPS-induced acute liver injury in mice

2.4.2.1. In vivo experimental protocols. For the in vivo experiments, we used a well-characterized sepsis model, wherein mice were stimulated with LPS to induce acute liver injury [10]. The model was established through an intraperitoneal (IP) injection of LPS (25 mg/kg, *E. coli* O127:B8; Sigma-Aldrich). Treatment groups were as follows: vehicle (0.9 % NaCl) (0.5 mL, IP), vehicle (0.9 % NaCl) plus EEXOs (1×10^9 particles per mouse, IP), LPS, LPS plus EEXOs, and LPS plus let-7i-5p inhibitor-treated EEXOs (1×10^9 particles per mouse, IP). Two consecutive doses of EEXOs or let-7i-5p inhibitor-treated EEXOs were administered at 2 and 26 h after LPS or PBS injection. The mice were closely monitored throughout the study.

The LPS dose (25 mg/kg) was selected based on previous studies demonstrating its ability to induce substantial injury in vital organs, including the lungs and liver [10,11]. The doses of EEXOs and let-7i-5p

inhibitor-treated EEXOs (1×10^9 particles per mouse) were likewise chosen according to earlier findings showing that EEXOs at this dose effectively mitigate LPS-induced acute lung injury, whereas let-7i-5p inhibitor-treated EEXOs at the same dose abolish this therapeutic effect [11]. Furthermore, our prior results indicated that EEXOs at 1×10^9 particles per mouse, but not at 1×10^8 particles per mouse, significantly improved 48-h survival in LPS-treated male mice (100 % vs. 50 %) [11]. In addition, intraperitoneal administration of EEXOs led to marked biodistribution to the lungs, liver, and kidneys, with lung accumulation peaking at 2 h, liver accumulation at 24–48 h, and renal accumulation between 2 and 48 h post-injection [11].

2.4.2.2. Liver injury analysis. At 48 h after LPS or normal saline injection, the mice that survived were anesthetized with zoletil (40 mg/kg, IP) and xylazine (10 mg/kg, IP) and then euthanized through decapitation. A midline laparotomy was performed, and blood samples were collected through aortic puncture. Blood was centrifuged at 2000 $\times$ g for 10 min, and plasma supernatants were collected and stored at -20°C for later analysis. The right and median liver lobes were snap-frozen in liquid nitrogen and stored at -80°C . The left lobe was fixed in 10 % formalin (Sigma-Aldrich) for histological analysis. The caudate lobe was freshly excised, weighed, dried at 80°C for 24 h, and reweighed to calculate the wet-to-dry ratio as a measure of liver water content [10].

Plasma levels of aspartate transaminase (AST) and alanine transaminase (ALT) were measured using a Vitros 750 autoanalyzer (Johnson & Johnson, New Brunswick, NJ, USA) to examine liver injury [10]. Formalin-fixed liver tissues were embedded in paraffin, sectioned, and stained with hematoxylin and eosin (Sigma-Aldrich). Histological characteristics of liver injury were evaluated under a light microscope (MoticEasyScan Pro 6; Motic Asia, Hong Kong). Liver injury was scored on the basis of the following histological characteristics: hepatic sinusoid congestion (0 = none, 1 = minimal, 2 = mild, 3 = moderate, and 4 = severe), cytoplasmic vacuolization (0 = none, 1 = minimal, 2 = mild, 3 = moderate, and 4 = severe), and parenchymal necrosis (0 = none, 1 = single-cell necrosis, 2 $\leq$ 30 %, 3 $\leq$ 60 %, and 4 $\geq$ 60 %). These scores were summed to determine the cumulative liver injury score (Suzuki's liver injury score) [36].

2.4.2.3. Effects of let-7i-5p on TLR4 and its downstream signaling components in the liver tissues of LPS-stimulated mice

2.4.2.3.1. Measurement of TLR4 mRNA levels in liver tissues. TLR4 mRNA levels in liver tissues were quantified through qPCR, following established protocols [10,37]. Snap-frozen liver tissue samples were thawed and processed for total RNA extraction [10]. After cDNA synthesis, qPCR was performed using standard procedures, with GAPDH as the internal standard [37]. Relative TLR4 mRNA levels were determined from the amplification data.

2.4.2.3.2. Measurement of TLR4 protein levels in liver tissues. TLR4 protein levels in liver tissues were measured through immunohistochemical staining, following a previously described protocol [11]. Formalin-fixed liver samples were sectioned and incubated with a primary antibody against TLR4 (1:200 dilution; IR324-1020; iReal Technology). For each sample, six randomly selected fields were imaged using a light microscope (MoticEasyScan Pro 6). Staining intensity was quantified using ImageJ, and the mean density across fields was calculated for analysis.

2.4.2.3.3. Measurement of the mRNA levels of TLR4's downstream signaling components in liver tissues. The mRNA levels of MyD88, TRAF6, NF- κ B, TNF- α , and IL-6 were quantified through qPCR, using GAPDH as the internal standard. The same RNA samples extracted from snap-frozen liver tissues were used. After cDNA synthesis, qPCR was performed following established protocols [37], and transcript levels were determined from the amplification curves.

2.5. Statistical analysis

Data are presented as mean ± standard deviation values. Intergroup differences were analyzed using one-way analysis of variance, followed by Tukey's post hoc test. A *p* value of <0.05 was considered statistically significant. All statistical analyses were performed using Prism (version 9; GraphPad Software, Boston, MA, USA).

3. Results

3.1. Characteristics of EEXOs

3.1.1. Morphology, size distribution, and exosome markers of EEXOs

TEM confirmed that EEXOs exhibited a double-layered, cup-shaped morphology (Fig. 1A). Nanoparticle tracking analysis revealed a particle size distribution of 88–189 nm, peaking at 106 nm (Fig. 1B). Immunoblotting confirmed the presence of the characteristic exosome markers ALIX, CD63, and CD9 on EEXOs (Fig. 1C).

3.1.2. Let-7i-5p concentration in EEXOs

ddPCR analysis revealed that in a 20-μL reaction, BEXOs contained 702.67 ± 189.41 copies of let-7i-5p, whereas EEXOs contained 8273.33 ± 1584.21 copies (Fig. 1D). Thus, the concentration of let-7i-5p was approximately an 11-fold higher in EEXOs than in BEXOs (*p* = 0.001).

3.2. Potential targets of let-7i-5p

Fig. 2 presents the bioinformatic analysis results from four microRNA target prediction databases. The initial screening identified 46 potential targets of let-7i-5p, as depicted in the Venn diagram. These targets were subsequently filtered focusing on inflammation-related molecules. TLR4 emerged as a key therapeutic target because of its essential role in detecting pathogens, activating inflammatory pathways, and modulating immune responses [20–22].

A binding site for let-7i-5p was identified within the murine TLR4 3'UTR sequence, characterized by Watson–Crick base pairing (|) and G•U wobble base pairing (:) [18].

	8	7	6	5	4	3	2	1
Let-7i-5p	G	A	U	G	G	A	G	U
	:					-		
Binding site	T	A	T	A	C	C	T	A
(177–184)	T	A	T	G	A	C	C	T
								CTCGGG

3.3. Results of in vitro experiments

3.3.1. Optimal doses of EEXOs and LPS for in vitro experiments

The viability of RAW264.7 cells treated with EEXOs (1×10^8 or 1×10^9 particles/mL) or LPS (50 or 100 ng/mL) was similar to that of PBS-treated cells (*p* = 0.503, *p* = 0.257, *p* = 0.067, and *p* = 0.223, respectively). By contrast, treatment with 500 ng/mL LPS significantly reduced viability compared with that of PBS-treated cells (*p* < 0.001; Fig. 3A). On the basis of these findings, we used the 100 ng/mL concentration of LPS in subsequent experiments.

To determine the optimal EEXO dose for the in vitro experiments, RAW264.7 cells were treated with PBS, LPS (100 ng/mL), or LPS plus EEXOs (1×10^8 or 1×10^9 particles/mL). Enzyme-linked immunosorbent assay revealed that IL-6 levels were significantly higher in LPS-treated cells than in PBS-treated cells (*p* < 0.001; Fig. 3B). IL-6 levels were significantly lower in cells treated with LPS plus EEXOs at 1×10^9 particles/mL than in those treated with LPS alone (*p* = 0.010; Fig. 3B). However, no significant difference in IL-6 level was observed between cells treated with LPS plus EEXOs at 1×10^8 particles/mL and those treated with LPS alone (*p* = 0.304; Fig. 3B). These findings confirmed the anti-inflammatory effect of EEXOs at 1×10^9 particles/mL in LPS-stimulated RAW264.7 cells.

Viability was similar between cells treated with LPS (100 ng/mL) alone and those treated with LPS plus EEXOs (1×10^9 particles/mL; *p* = 0.473; Fig. 3C). These findings confirmed that EEXOs at 1×10^9 particles/mL did not adversely affect cell viability, regardless of LPS stimulation. On the basis of these results, the 1×10^9 particles/mL dose of EEXOs was used in subsequent in vitro experiments.

3.3.2. In vitro biobehavior of EEXOs

3.3.2.1. EEXO uptake by RAW264.7 cells. Representative confocal microscopy images are presented in Fig. 4A & 4B. The nuclei, EEXOs, and plasma membranes were labeled with Hoechst 33342 (blue fluorescence), ExoGlow-Protein (green fluorescence), and PlasMem Bright dye (red fluorescence), respectively. Fig. 4A depicts the interaction between the green fluorescence of EEXOs and the red fluorescence of the plasma membrane, indicating EEXO uptake by RAW264.7 cells after 1 h of incubation. The merged image in Fig. 4A further confirms contact between EEXOs and the plasma membrane at this time point.

At 12 h of incubation (Fig. 4B), EEXOs were internalized by RAW264.7 cells, as indicated by overlapping green and red fluorescence signals. The merged image in Fig. 4B confirmed the cytoplasmic

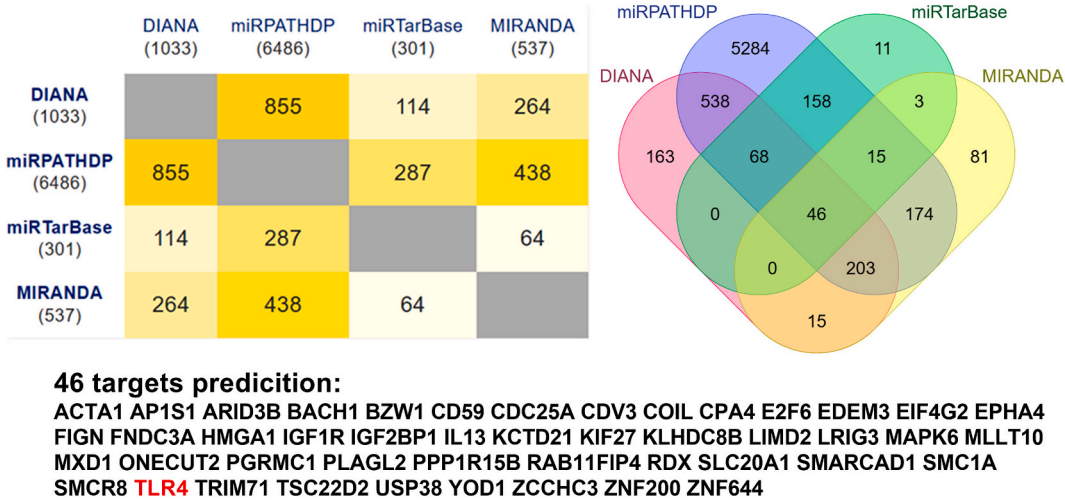


Fig. 2. Bioinformatic analysis predicts toll-like receptor 4 (TLR4) as a target of microRNA let-7i-5p. The mRNA targets of let-7i-5p were predicted using four algorithms: DIANA-microT 2023, miRPATHDP, miRTarBase, and MiRanda. Bioinformatics tools (<https://molbiotools.com>) were used to process the data, initially identifying 46 overlapping targets, as shown in the Venn diagram. Further screening of inflammation-related molecules identified TLR4 as a key target.

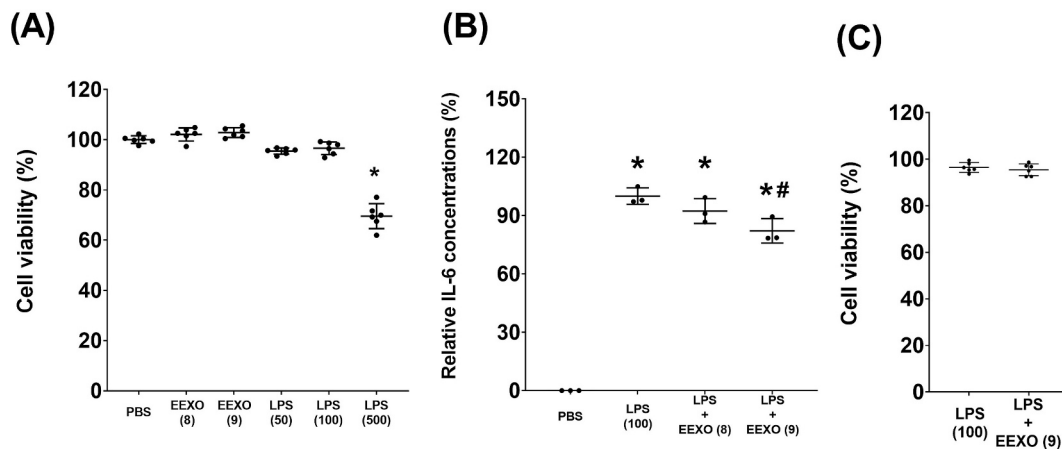


Fig. 3. Determination of optimal doses of engineered exosomes enriched with microRNA let-7i-5p (EEXO) and lipopolysaccharide (LPS) for the in vitro experiments. (A) Cell viability in RAW264.7 cells, measured using the cell counting kit-8 assay. PBS: RAW264.7 cells treated with phosphate-buffered saline (PBS) ($n = 6$). EEXO (8) and EEXO (9): RAW264.7 cells treated with PBS plus 1×10^8 and 1×10^9 particles/mL of EEXO, respectively. LPS (50), LPS (100), and LPS (500): RAW264.7 cells treated with 50, 100, and 500 ng/mL of LPS, respectively ($n = 6$ per group). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the PBS group. (B) Relative interleukin-6 (IL-6) concentrations in RAW264.7 cells, measured using enzyme-linked immunosorbent assay. PBS: RAW264.7 cells treated with PBS ($n = 3$). LPS (100): RAW264.7 cells treated with 100 ng/mL of LPS ($n = 3$). LPS + EEXO (8) and LPS + EEXO (9): RAW264.7 cells treated with LPS plus EEXO at 1×10^8 particles/mL and 1×10^9 particles/mL, respectively ($n = 3$ per group). IL-6 concentration in the LPS group was used for comparison. Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the PBS group. # $p < 0.05$, versus the LPS group. (C) Cell viability in RAW264.7 cells treated with 100 ng/mL of LPS. LPS + EEXO (9): RAW264.7 cells treated with LPS plus EEXO at 1×10^9 particles/mL ($n = 5$ per group). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test.

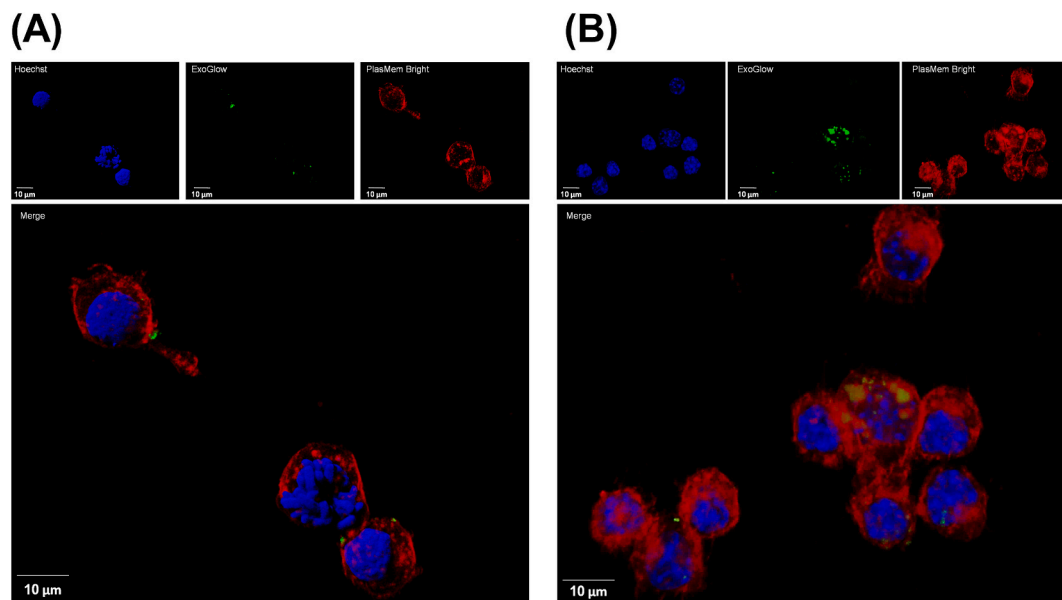


Fig. 4. Representative confocal microscopy images showing the uptake of engineered exosomes enriched with microRNA let-7i-5p (EEXO) by RAW264.7 cells. EEXO were labeled with ExoGlow-Protein (green fluorescence), while the plasma membrane and nuclei of RAW264.7 cells were stained with Plasmem Bright (red fluorescence) and Hoechst 33342 (blue fluorescence), respectively. (A) EEXO-cell interaction after 1 h of incubation. (B) EEXO-cell interaction after 12 h of incubation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

localization of EEXOs in RAW264.7 cells after 12 h of incubation.

3.3.2.2. EEXO-mediated delivery of let-7i-5p to target cells. RAW264.7 cells were treated with PBS or EEXOs (1×10^9 particles/mL; $n = 5$ per group). After 12 h of incubation, the cells were harvested, and intracellular let-7i-5p concentrations were quantified through qPCR. EEXO-treated cells exhibited a 3-fold higher let-7i-5p expression than did PBS-treated cells ($p < 0.001$; Fig. 5A). These results confirm that EEXOs efficiently delivered let-7i-5p to target cells.

3.3.2.3. Binding of let-7i-5p to TLR4 mRNA. A luciferase assay was conducted to validate the interaction between let-7i-5p and TLR4 mRNA. Plasmids containing pMirTarget, pMirTLR4, or pMirTLR4mut were successfully transfected into RAW264.7 cells. The transfected cells were then treated with PBS or EEXOs (1×10^9 particles/mL), resulting in six groups ($n = 6$ per group): PBS plus vector (Vector), EEXOs plus vector (Vector_EEXO), PBS plus wild-type TLR4 (WT), EEXOs plus wild-type TLR4 (WT_EEXO), PBS plus mutant TLR4 (MUT), and EEXOs plus mutant TLR4 (MUT_EEXO).

As shown in Fig. 5B, luminescent signal intensity in the WT_EEXO

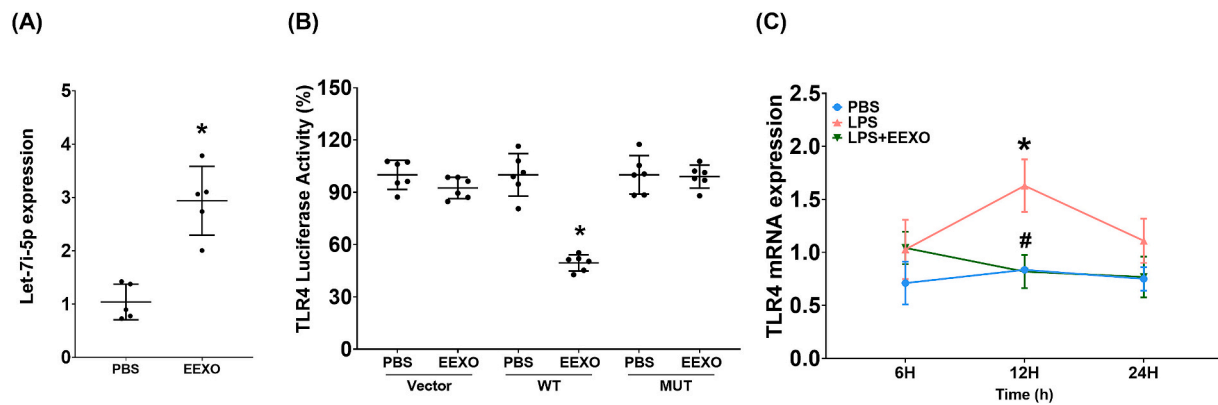


Fig. 5. Effects of microRNA let-7i-5p, delivered via engineered exosomes enriched with microRNA let-7i-5p (EEXO), on inhibiting toll-like receptor 4 (TLR4) mRNA expression. (A) Effects of let-7i-5p delivery to RAW264.7 cells, mediated by engineered exosomes enriched with let-7i-5p (EEXO). Let-7i-5p concentrations in RAW264.7 cells treated with phosphate-buffered saline (PBS, PBS group) and EEXO at 1×10^9 particles/mL (EEXO group) were measured using reverse transcription-quantitative polymerase chain reaction ($n = 5$ per group). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, the EEXO group versus the PBS group. (B) Binding assay of let-7i-5p and TLR4 mRNA using a luciferase activity assay. Plasmids containing the control vector (pMirTarget), the 3'UTR sequence of wild-type TLR4 mRNA (pMirTLR4), or the 3'UTR sequence of mutant TLR4 mRNA (pMirTLR4mut) were transfected into RAW264.7 cells. The transfected cells were treated with either PBS or EEXO (1×10^9 particles/mL), forming six groups ($n = 6$ per group): PBS plus vector (Vector), EEXO plus vector (Vector_EEXO), PBS plus wild-type TLR4 (WT), EEXO plus wild-type TLR4 (WT_EEXO), PBS plus mutant TLR4 (MUT), and EEXO plus mutant TLR4 (MUT_EEXO). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the WT group. (C) Effects of let-7i-5p on reducing TLR4 mRNA upregulation in lipopolysaccharide (LPS)-stimulated RAW264.7 cells. RAW264.7 cells were treated with PBS (PBS group), LPS (100 ng/mL, LPS group), or LPS plus EEXO (1×10^9 particles/mL, LPS + EEXO group). TLR4 mRNA concentrations were measured at 6, 12, and 24 h after LPS treatment ($n = 3$ per time point from each group), using reverse transcription-quantitative polymerase chain reaction with glyceraldehyde-3-phosphate dehydrogenase as the internal control. Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the PBS group. # $p < 0.05$, versus the LPS group.

group decreased to 49.5 % compared with that in the WT group ($p < 0.001$). By contrast, luminescent signal intensity in the MUT_EEXO group was similar to that in the MUT, Vector, and Vector_EEXO groups (all $p > 0.05$). These results suggest that let-7i-5p directly binds to the 3'UTR of TLR4 mRNA, thereby inhibiting TLR4 translation and luciferase expression.

3.3.3. Let-7i-5p inhibits LPS-induced TLR4-mediated signaling in vitro

3.3.3.1. Let-7i-5p inhibits the LPS-induced upregulation of TLR4 mRNA in RAW264.7 cells. To investigate the effects of let-7i-5p on LPS-induced TLR4 mRNA expression, RAW264.7 cells were treated with PBS, LPS (100 ng/mL), or LPS plus EEXOs (1×10^9 particles/mL). These groups were designated as the PBS, LPS and LPS + EEXO, respectively. In the LPS + EEXO group, EEXOs were added 1 h before LPS challenge. Cell lysates for qPCR were collected at 6, 12, and 24 h after treatment ($n = 3$ per time point).

Fig. 5C presents the qPCR results. The mRNA level of TLR4 was significantly higher in the LPS group than in the PBS group ($p < 0.001$) at 12 h after LPS challenge, whereas the levels at 6 and 24 h after LPS challenge did not differ significantly between the two groups. Notably, the mRNA level of TLR4 at 12 h after LPS challenge was significantly lower in the LPS + EEXO group than in the LPS group ($p < 0.001$). At 6 and 24 h after LPS challenge, no significant differences were noted between the LPS + EEXO and LPS groups. These findings indicated that let-7i-5p suppressed the LPS-induced upregulation of TLR4 mRNA in RAW264.7 cells.

3.3.3.2. Let-7i-5p inhibits TLR4-mediated signaling in LPS-stimulated RAW264.7 cells. To investigate the effects of let-7i-5p on TLR4-mediated signaling cascades, RAW264.7 cells were treated with PBS, EEXOs (1×10^9 particles/mL), LPS (100 ng/mL), LPS plus EEXOs (1×10^9 particles/mL), and LPS plus let-7i-5p inhibitor-treated EEXOs (1×10^9 particles/mL). These groups were designated as PBS, EEXO, LPS, LPS + EEXO, and LPS + EEXO (i), respectively ($n = 5$ per group). In the LPS + EEXO and LPS + EEXO(i) groups, EEXOs or let-7i-5p

inhibitor-treated EEXOs were added 1 h before LPS challenge. Cell lysates were collected 12 h after LPS challenge, and mRNA levels of TLR4 and its downstream signaling components were quantified through qPCR.

As shown in Fig. 6, the mRNA levels of TLR4, MyD88, TRAF6, NF- κ B, TNF- α , and IL-6 were similar between the PBS and EEXO groups. By contrast, the mRNA levels of TLR4 ($p = 0.002$), MyD88 ($p < 0.001$), TRAF6 ($p = 0.008$), NF- κ B ($p < 0.001$), TNF- α ($p < 0.001$), and IL-6 ($p < 0.001$) were significantly higher in the LPS group than in the PBS group. Moreover, the mRNA levels of TLR4, MyD88, TRAF6, NF- κ B, TNF- α and IL-6 were significantly lower in the LPS + EEXO group than in the LPS group ($p < 0.001$, $p = 0.003$, $p = 0.004$, $p = 0.047$, $p < 0.001$, and $p = 0.004$, respectively). Furthermore, the mRNA levels of TLR4, MyD88, TRAF6, NF- κ B, TNF- α , and IL-6 were significantly higher in the LPS + EEXO(i) group than in the LPS + EEXO group ($p < 0.001$, $p = 0.012$, $p = 0.006$, $p = 0.001$, $p < 0.001$, and $p = 0.007$, respectively). Collectively, these findings indicated that let-7i-5p suppressed TLR4 upregulation and thereby downregulated TLR4-mediated signaling in LPS-stimulated RAW264.7 cells.

3.4. Results of in vivo experiments

3.4.1. Let-7i-5p mitigates LPS-induced liver injury in mice

To validate the in vitro findings, we conducted in vivo experiments by using an LPS-induced murine model of liver injury. Adult male mice received intraperitoneal injections of normal saline, EEXOs, LPS (25 mg/kg), LPS plus EEXOs, or LPS plus let-7i-5p inhibitor-treated EEXOs. These groups were designated as the Sham, EEXO, LPS, LPS + EEXO, and LPS + EEXO (i), respectively ($n = 5$ per group). Mice in the EEXO and LPS + EEXO groups received two consecutive intraperitoneal injections of EEXOs (1×10^9 particles per mouse) administered 2 and 26 h after LPS challenge, and mice in the LPS + EEXO(i) group received two equivalent doses of let-7i-5p inhibitor-treated EEXOs at the same time points. All animals were euthanized at 48 h after LPS challenge or at the corresponding time point for controls.

Fig. 7A presents representative hematoxylin-eosin-stained liver

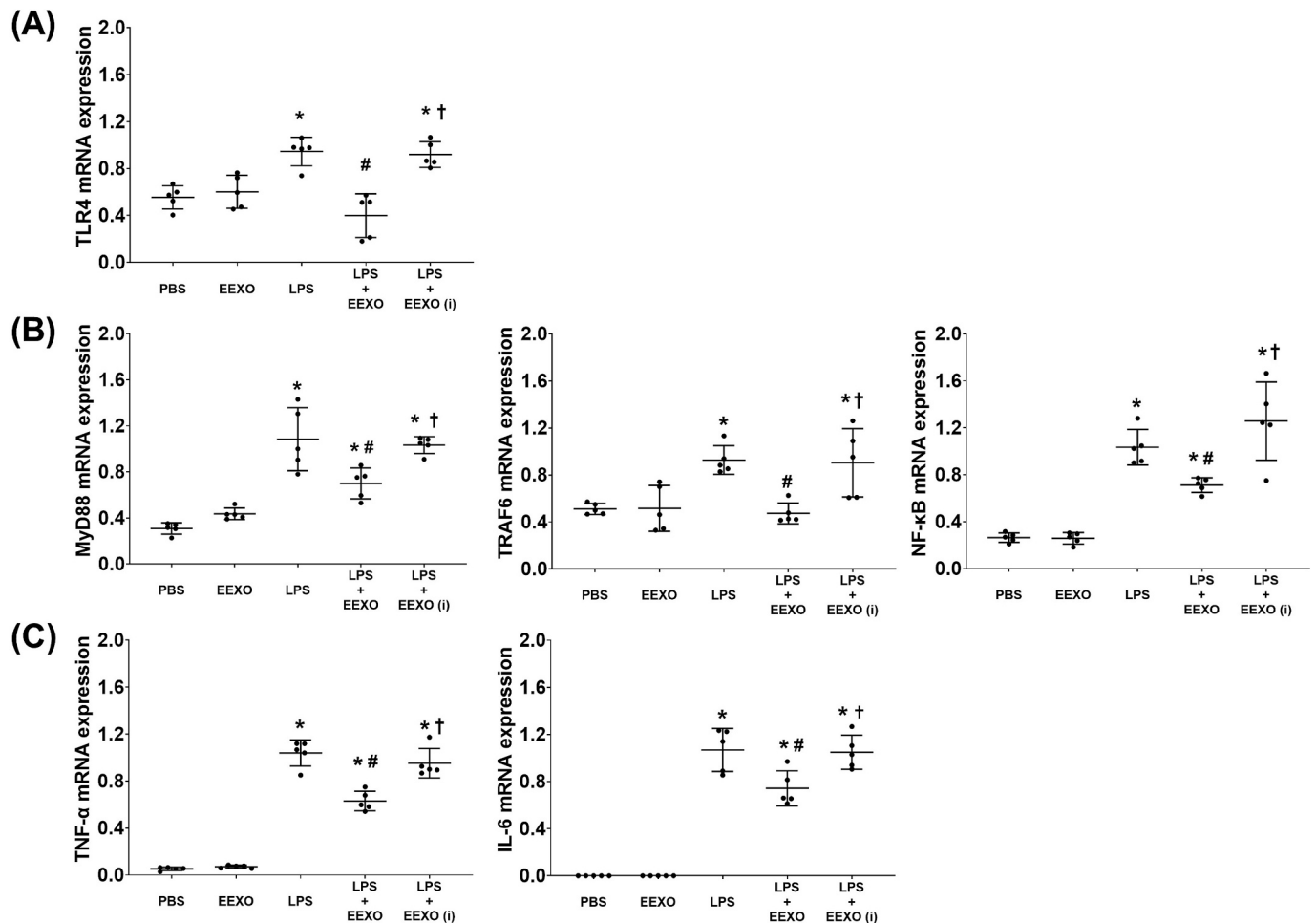


Fig. 6. Effects of microRNA let-7i-5p, delivered via engineered exosomes enriched with microRNA let-7i-5p (EEXO), on inhibiting toll-like receptor 4 (TLR4) and TLR4-mediated signaling pathways in lipopolysaccharide (LPS)-treated RAW264.7 cells. The mRNA concentrations of (A) TLR4, (B) myeloid differentiation primary response 88 (MyD88), tumor necrosis factor receptor-associated factor 6 (TRAF6), nuclear factor- κ B (NF- κ B), and (C) tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6) were measured using reverse transcription-quantitative polymerase chain reaction, with glyceraldehyde-3-phosphate dehydrogenase as the internal control. PBS and EEXO: RAW264.7 cells treated with phosphate-buffered saline (PBS) and EEXO (1×10^9 particles/mL), respectively ($n = 5$ per group). LPS, LPS + EEXO, and LPS + EEXO (i): RAW264.7 cells treated with LPS (100 ng/mL), LPS plus EEXO (1×10^9 particles/mL), or LPS plus EEXO pretreated with a let-7i-5p inhibitor (1×10^9 particles/mL), respectively ($n = 5$ per group). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the PBS group. # $p < 0.05$, versus the LPS group. † $p < 0.05$, the LPS + EEXO (i) group versus the LPS + EEXO group.

sections and corresponding Suzuki's liver injury scores. The Sham and EEXO groups exhibited no significant evidence of congestion, cytoplasmic vacuolization, or parenchymal necrosis. By contrast, significant liver injury was observed in the LPS, LPS + EEXO, and LPS + EEXO(i) groups. Notably, the severity of injury was lower in the LPS + EEXO group than in the LPS group.

Suzuki's liver injury scores supported the aforementioned observations. The Sham and EEXO groups had lower scores than did the other groups. The LPS group had a significantly higher score than did the Sham group ($p < 0.001$). The LPS + EEXO group had a significantly lower score than did the LPS group ($p < 0.001$), whereas the LPS + EEXO (i) group had a significantly higher score than did the LPS + EEXO group ($p = 0.002$). Biochemical analyses corroborated these results. Plasma AST and ALT levels (Fig. 7B) and the liver wet-to-dry ratio (Fig. 7C) followed the same pattern as Suzuki's scores. Collectively, these findings suggested that engineered exosome-mediated delivery of let-7i-5p mitigated LPS-induced acute liver injury in adult male mice and that the treatment of EEXOs with an let-7i-5p inhibitor abolished this therapeutic effect.

3.4.2. Let-7i-5p suppresses the LPS-induced upregulation of TLR4-mediated signaling in the liver tissues of adult male mice

Fig. 8A presents TLR4 mRNA levels in snap-frozen liver tissues and TLR4 protein expression in formalin-fixed tissues in the five experimental groups. TLR4 mRNA levels were lower in the Sham and EEXO groups than in the other groups. However, the levels were significantly higher in the LPS group than in the Sham group ($p < 0.001$). Furthermore, TLR4 mRNA levels were significantly lower in the LPS + EEXO group than in the LPS group ($p < 0.001$). The levels were significantly higher in the LPS + EEXO(i) group than in the LPS + EEXO group ($p = 0.019$). Consistent with these findings, immunohistochemical staining revealed strong TLR4 signals in the LPS and LPS + EEXO(i) groups but weak signals in the Sham, EEXO, and LPS + EEXO groups. Quantitative analysis of staining intensities closely paralleled the mRNA expression patterns.

mRNA expression data for MyD88, TRAF6, NF- κ B, TNF- α , and IL-6 in snap-frozen liver tissues across the five experimental groups are presented in Fig. 8B & 8C. The mRNA levels of MyD88, TRAF6, NF- κ B, TNF- α , and IL-6 were lower in the Sham and EEXO groups than in the other groups. However, these levels were significantly higher in the LPS group than in the Sham group ($p < 0.001$ for MyD88, TRAF6, NF- κ B, and TNF- α ; $p = 0.028$ for IL-6). The LPS + EEXO group had significantly lower

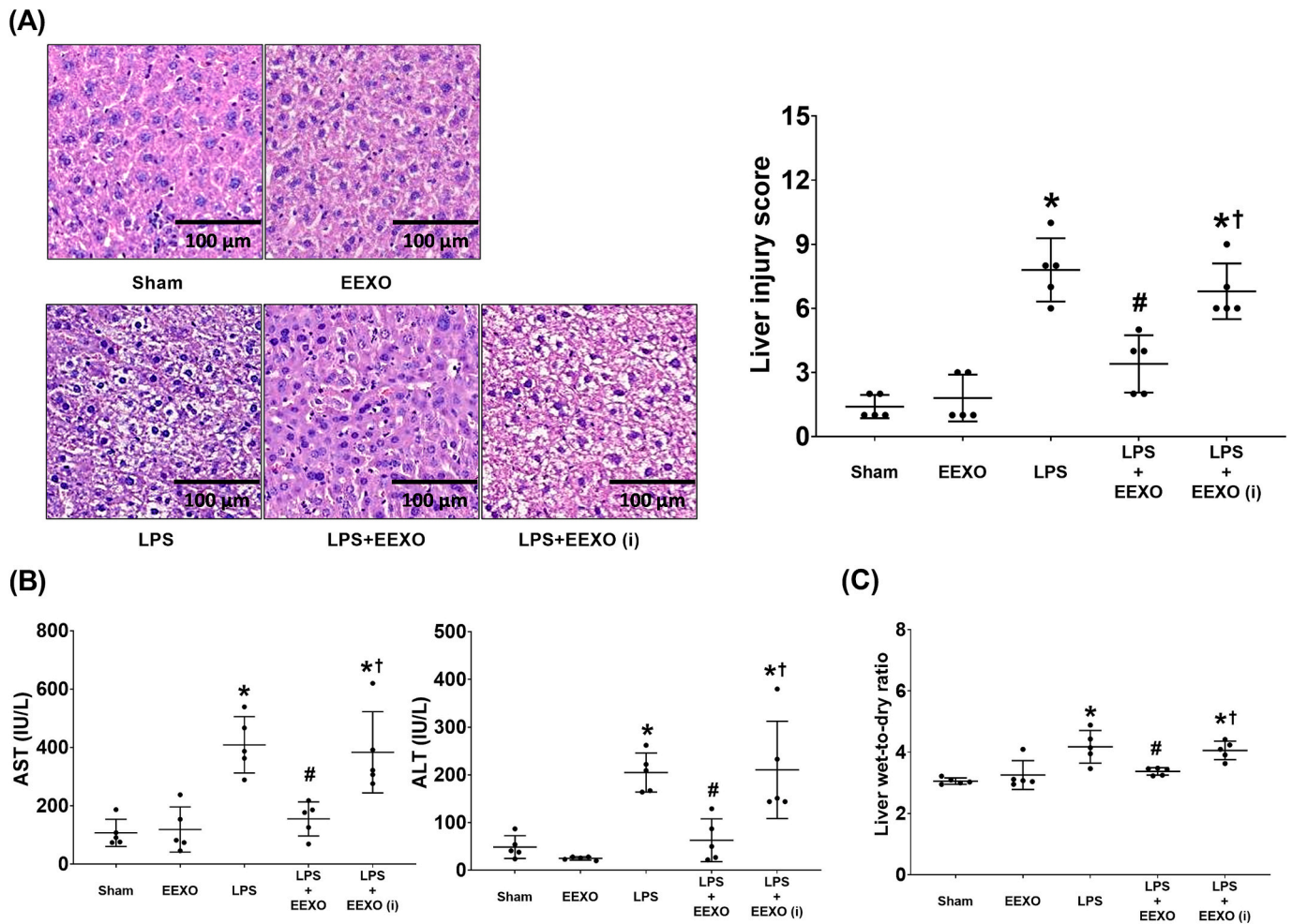


Fig. 7. Effects of microRNA let-7i-5p, delivered via engineered exosomes enriched with microRNA let-7i-5p (EEXO), in mitigating acute liver injury in lipopolysaccharide (LPS)-treated adult male C57B/L6 mice. (A) Representative microscopic images of liver tissues stained with hematoxylin-eosin (400 $\times$), along with corresponding liver injury scores assessed using Suzuki's liver injury scoring system. (B) Measurements of liver enzyme levels, including aspartate transaminase (AST) and alanine transaminase (ALT). (C) The liver wet-to-dry ratio. Sham and EEXO: Mice treated with normal saline and EEXO (1×10^9 particles per mouse), respectively ($n = 5$ per group). LPS, LPS + EEXO, and LPS + EEXO (i): mice treated with LPS (25 mg/kg), LPS plus EEXO (1×10^9 particles per mouse), or LPS plus EEXO pretreated with a let-7i-5p inhibitor (1×10^9 particles per mouse), respectively ($n = 5$ per group). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the PBS group. # $p < 0.05$, versus the LPS group. † $p < 0.05$, the LPS + EEXO (i) group versus the LPS + EEXO group.

mRNA levels of MyD88 ($p = 0.001$), TRAF6 ($p < 0.001$), NF- κ B ($p < 0.001$), TNF- α ($p < 0.001$), and IL-6 ($p = 0.037$) than did the LPS group. Furthermore, the LPS + EEXO(i) group had significantly higher mRNA levels of MyD88 ($p < 0.001$), TRAF6 ($p < 0.001$), NF- κ B ($p = 0.006$), TNF- α ($p = 0.006$), and IL-6 ($p = 0.021$) than did the LPS + EEXO group.

The findings confirmed that engineered exosome-mediated delivery of let-7i-5p suppressed the LPS-induced upregulation of TLR4, its downstream signaling components, and proinflammatory cytokines in the liver tissues of adult male mice.

4. Discussion

4.1. Therapeutic potential of let-7i-5p delivery via engineered exosomes

To enable efficient delivery of let-7i-5p, we prepared engineered exosomes by genetically modifying RAW264.7 cells to overexpress this microRNA. Engineered exosomes enriched with therapeutic microRNAs, such as let-7i-5p, represent a novel and promising strategy for the targeted treatment of inflammatory diseases. As naturally secreted nanovesicles, exosomes possess several intrinsic advantages pertaining to the delivery of small molecules—for example, they are biocompatible,

nonimmunogenic, and capable of crossing biological barriers [14–17]. Thus, engineered exosomes carrying therapeutic microRNAs provide a cell-free, precision-targeted approach with considerable advantages over conventional gene- or drug-based therapies [14–17].

The present study demonstrated that genetic modification of RAW264.7 cells to overexpress let-7i-5p significantly increased the therapeutic content of engineered exosomes, resulting in an approximately 11-fold increase in let-7i-5p concentration compared with that in BEXOs. The engineered exosomes preserved key exosomal characteristics, such as morphology, particle size distribution, and classical markers (CD9, CD63, and ALIX) expression, which confirmed their integrity and suitability for therapeutic use. EEXOs were determined to be safe for RAW264.7 cells. The uptake of EEXOs by these cells resulted in an approximately 3-fold increase in the intracellular concentration of let-7i-5p, demonstrating efficient delivery. The increase let-7i-5p concentration significantly alleviated inflammation both in vitro and in vivo. Collectively, these findings highlight engineered exosome-mediated let-7i-5p delivery as a promising therapeutic approach for inflammation-related diseases, such as sepsis.

This study confirmed TLR4 as a direct target of let-7i-5p, representing the first demonstration of this interaction. TLR4 is a crucial

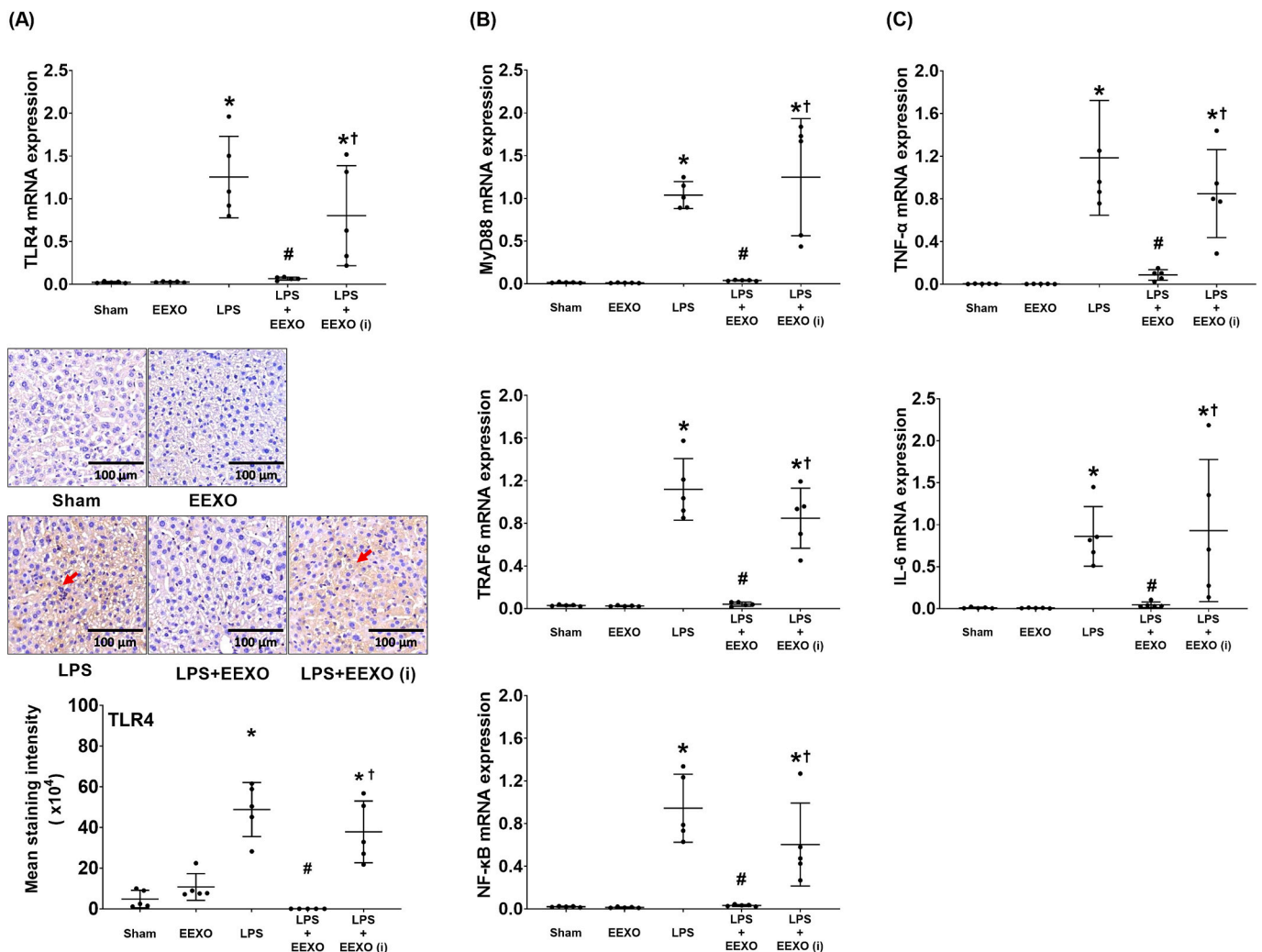


Fig. 8. Effects of microRNA let-7i-5p, delivered via engineered exosomes enriched with microRNA let-7i-5p (EEXO), on inhibiting toll-like receptor 4 (TLR4) and TLR4-mediated signaling pathways in liver tissues of lipopolysaccharide (LPS)-treated adult male C57B/L6 mice. (A) The TLR4 mRNA levels in liver tissues, measured by reverse transcription-quantitative polymerase chain reaction (qPCR) with glyceraldehyde-3-phosphate dehydrogenase (GAPDH) as the internal control, are shown. Representative photomicrographs and signal intensities from an immunohistochemistry staining assay are also presented to illustrate TLR4 expression (indicated by the red arrow) in liver tissues. (B) The mRNA levels of myeloid differentiation primary response 88 (MyD88), tumor necrosis factor receptor-associated factor 6 (TRAF6), and nuclear factor-κB (NF-κB) in liver tissues, measured by qPCR with GAPDH as the internal control, are shown. (C) The mRNA levels of tumor necrosis factor-α (TNF-α) and interleukin-6 (IL-6) in liver tissues, also measured by qPCR with GAPDH as the internal control, are shown. Sham and EEXO: Mice treated with normal saline and EEXO (1×10^9 particles per mouse), respectively ($n = 5$ per group). LPS, LPS + EEXO, and LPS + EEXO (i): mice treated with LPS (25 mg/kg), LPS plus EEXO (1×10^9 particles per mouse), or LPS plus EEXO pretreated with a let-7i-5p inhibitor (1×10^9 particles per mouse), respectively ($n = 5$ per group). Data are presented as mean $\pm$ standard deviations and analyzed by the one-way analysis of variance followed by Tukey's post hoc test. * $p < 0.05$, versus the PBS group. # $p < 0.05$, versus the LPS group. † $p < 0.05$, the LPS + EEXO (i) group versus the LPS + EEXO group. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

pattern recognition receptor that mediates host responses to bacterial endotoxins such as LPS, primarily through the activation of the MyD88/TRAF6/NF-κB pathway and subsequent induction of proinflammatory cytokines, such as TNF-α and IL-6, which contribute to systemic inflammation and tissue injury [20,21]. Our in vitro and in vivo findings confirmed that let-7i-5p suppressed LPS-induced TLR4 upregulation, subsequently inhibiting the downstream MyD88/TRAF6/NF-κB pathway and proinflammatory cytokines (TNF-α and IL-6). Furthermore, in vivo experiments demonstrated that let-7i-5p mitigated LPS-induced liver injury in mice. These effects and the underlying molecular mechanisms are summarized in the graphical abstract. Collectively, the results suggest that let-7i-5p exerts potent anti-inflammatory effects by inhibiting TLR4. Thus, this microRNA holds promise as a novel therapeutic candidate for TLR4-driven inflammatory diseases, particularly sepsis. Given the vital role of TLR4 in regulating inflammatory

pathways [20,22,38] and the lack of effective therapies for sepsis [39], our findings have major clinical implications and warrant further exploration.

4.2. This study versus the literature

Our findings complement and extend recent advances in engineered exosome-microRNA therapies for inflammatory diseases. For example, a study demonstrated the efficacy of Wharton's jelly MSC-derived exosomes enriched with let-7a-5p (through cellular nanoporation/electroporation) in a rat model of hyperoxia-induced acute lung injury [40]. These exosomes improved lung function and reduced collagen deposition after intravenous administration; the effects were mediated primarily through antifibrotic and immunomodulatory mechanisms, such as the downregulation of SMAD2/3 signaling, promotion of M2-like

macrophage polarization, and upregulation of IL-10. Unlike the aforementioned study [40], we evaluated the efficacy of macrophage-derived EEXOs in an LPS-induced acute liver injury model. Our findings highlighted the EEXO-mediated suppression of TLR4/MyD88/TRAF6/NF- κ B signaling as the principal therapeutic mechanism. We previously demonstrated that this approach could mitigate LPS-induced acute lung injury in mice [11]. This finding supports the broader applicability of let-7i-5p-based exosome therapy across multiple organ systems affected by sepsis.

Together, the findings underscore both commonalities and differences between the aforementioned strategy [40] and our strategy. Engineered exosomes can confer protection through different therapeutic mechanisms—proresolution and antifibrotic effects (let-7a-5p, SMAD2/3, and IL-10) or anti-pathogen-sensing and anti-inflammatory effects (let-7i-5p and TLR4–NF- κ B). The strategies further differ in exosome source (MSC vs. macrophage), engineering method (nanoporation vs. stable overexpression), disease trigger (hyperoxia vs. endotoxemia), and administration route (intravenous vs. intraperitoneal). Collectively, the findings suggest that tailoring microRNA cargo and exosome origin to the underlying pathology—hyperoxia-induced fibrotic remodeling in the lungs versus endotoxin-driven inflammation in the liver—can maximize therapeutic efficacy. This rationale highlights the feasibility of combination strategies; for example, let-7i-5p may be combined with antifibrotic cargo for the treatment of persistent acute respiratory distress syndrome complicated by secondary liver injury.

Our results reinforce the growing body of evidence suggesting that exosome- and microRNA-based therapies can ameliorate sepsis and associated organ injury. MSC-derived exosomes exert their protective effects by modulating immune responses; for example, human umbilical cord MSC-derived exosomes mitigated acute lung injury by promoting macrophage polarization [41], and placenta MSC-derived exosomes alleviated systemic inflammation through abundant let-7 family members [10,11,42]. Other microRNAs, such as miR-146a and miR-21, suppress NF- κ B signaling [43,44], whereas miR-206-3p mitigates LPS-induced acute lung injury by targeting the TLR4/nucleotide-binding oligomerization domain leucine-rich repeat and pyrin domain containing 3 axis [45]. Notably, let-7i-5p is the first microRNA that has been confirmed to directly regulate TLR4. By demonstrating let-7i-5p binding to the TLR4 3'UTR, the present study elucidated a novel post-transcriptional mechanism, thereby expanding the therapeutic repertoire of microRNAs for sepsis and related inflammatory diseases.

4.3. Mechanistic insights into TLR4 mRNA dynamics

Activation of TLR4 by LPS initiates a tightly orchestrated signaling cascade that coordinates the innate immune response. Upon recognition, LPS binds to LPS-binding protein (LBP) and is transferred to the TLR4 receptor complex via the co-receptor CD14 [20]. Myeloid differentiation factor 2 (MD-2) facilitates the dimerization of TLR4/MD-2, a key step that triggers receptor internalization through endocytosis [46]. This dimerization event activates the MyD88/TRAF6/NF- κ B pathway, leading to the production of proinflammatory cytokines such as TNF- α and IL-6 [47]. As endocytosis proceeds, MyD88-dependent NF- κ B signaling is downregulated [47,48], and internalized receptors are directed either toward lysosomal degradation or recycling to the plasma membrane [49]. NF- κ B can further enhance TLR4 transcription, replenishing receptor levels and maintaining signaling capacity [38].

Beyond transcriptional regulation, posttranscriptional control of TLR4 mRNA stability represents another layer of immune modulation. The 3'UTR of TLR4 contains AU-rich elements (AREs), rendering the transcript inherently unstable and susceptible to degradation by ARE-binding proteins [50]. During LPS stimulation, however, stabilizing factors such as HuR can bind these regions to protect the transcript from decay, thereby transiently sustaining receptor expression [51]. This interplay between degradation and stabilization enables a rapid but self-

limiting inflammatory response.

Consistent with these mechanisms, our results showed that TLR4 mRNA expression peaked at 12 h following LPS exposure, with no significant changes at 6 or 24 h (Fig. 5C). This transient elevation likely reflects the combined influence of NF- κ B-driven transcriptional activation and HuR-mediated mRNA stabilization, followed by compensatory receptor internalization, degradation, and recycling. The short half-life of TLR4 mRNA further contributes to this transient expression pattern, suggesting that the 12-h peak represents a critical regulatory juncture within the self-limiting dynamics of TLR4 signaling [49–51].

We further identified let-7i-5p as a direct posttranscriptional regulator of TLR4. Luciferase reporter assays confirmed specific binding of let-7i-5p to the TLR4 3'UTR, resulting in reduced reporter activity, which was abolished when the binding site was mutated. Functionally, overexpression of let-7i-5p suppressed LPS-induced TLR4 mRNA upregulation, likely by promoting transcript degradation. These findings suggest that let-7i-5p counteracts the stabilizing effect of LPS-activated RNA-binding proteins such as HuR, restoring physiological mRNA turnover and preventing sustained receptor accumulation.

The downstream consequences of this regulation were evident in both in vitro and in vivo systems. In LPS-stimulated RAW264.7 cells, let-7i-5p reduced TLR4 expression, suppressed MyD88/TRAF6/NF- κ B activation, and diminished TNF- α and IL-6 secretion. In mouse models, administration of let-7i-5p-enriched EEXOs markedly alleviated LPS-induced acute liver injury, as reflected by lower serum AST and ALT levels and reduced liver wet-to-dry ratios, indicating attenuated inflammation and improved organ integrity. Conversely, inhibition of let-7i-5p exacerbated tissue injury, accompanied by elevated TLR4 expression, enhanced NF- κ B activation, and increased cytokine production.

Taken together, these findings identify let-7i-5p as a key negative feedback regulator of TLR4 signaling. By promoting degradation of TLR4 mRNA and dampening NF- κ B-mediated cytokine expression, let-7i-5p restrains excessive inflammation and protects against tissue injury. These results highlight the therapeutic potential of let-7i-5p-based interventions for inflammatory disorders driven by aberrant TLR4 activation.

4.4. Advantages of using a nonstem cell source for exosome production

A key strength of our approach is the use of a nonstem cell source for exosome production. Although MSC-derived exosomes have attracted considerable attention for their therapeutic potential, concerns remain regarding batch-to-batch variability, scalability, and potential oncogenic risks associated with the use of stem cell sources [52]. By contrast, the use of a well-characterized and stable cell line such as RAW264.7 provides several practical advantages: ease of genetic manipulation, reproducible exosome yield and cargo content, and fewer ethical and regulatory challenges than those with stem cell-derived products. In addition, macrophage-derived exosomes may confer further benefits in inflammatory contexts because of their intrinsic homing capacity to inflamed tissues [53]. Collectively, these attributes make RAW264.7 macrophage-derived engineered exosomes a scalable, controllable, and safe platform for microRNA delivery, supporting their translational potential for treating inflammation-associated diseases.

4.5. Translational significance and clinical implications

At the clinical level, TLR4 has long been considered a promising therapeutic target in sepsis and related inflammatory diseases; however, direct pharmacologic inhibition has yielded disappointing results. Eritoran (E5564), a synthetic lipid A analog that competitively blocks LPS binding to the TLR4/MD-2 complex, was evaluated in the Phase III ACCESS trial but failed to reduce 28-day mortality in nearly 2000 patients with sepsis [54]. Similarly, TAK-242 (Resatorvid), a small-molecule cyclohexene derivative that binds to the intracellular TIR

domain of TLR4 and prevents adaptor protein recruitment, exhibited a favorable safety profile in Phase I studies but lacked efficacy in Phase II trials [55]. These outcomes highlight the limitations of global TLR4 blockade, which may impair essential host defenses while failing to adequately restrain hyperinflammation.

Exosomal delivery of let-7i-5p offers a relatively refined strategy by selectively modulating TLR4 expression at the posttranscriptional level rather than abolishing signaling entirely. This fine-tuning may preserve antimicrobial immunity while reducing cytokine-driven injury, potentially overcoming the barriers that limited previous TLR4-targeted therapies. These findings highlight the need for additional investigations into engineered exosome-microRNA therapies for sepsis-associated organ injury.

In addition to facilitating the development of sepsis, TLR4 activation contributes to the pathogenesis of several chronic inflammatory diseases, including nonalcoholic fatty liver disease and autoimmune disorders [56,57]. Therefore, the therapeutic potential of exosomal let-7i-5p may extend to a broader spectrum of conditions characterized by aberrant TLR4 signaling. Supporting this concept, studies have reported that placenta MSC-derived exosomes exert protective effects in these settings, with let-7 family members emerging as key mediators. Further research is required to establish the efficacy, durability, and safety of engineered exosomal let-7i-5p therapy across clinical contexts.

4.6. Study limitations

This study has several limitations. First, we investigated the effects of EEXO-mediated let-7i-5p delivery in a short timeframe—24 h in vitro (RAW264.7 cells) and 48 h in vivo (acute liver injury model). Longer-term studies are required to determine the durability of therapeutic efficacy and potential toxicity of EEXOs. In addition, we tested only a single therapeutic protocol for mitigating LPS-induced liver injury in mice. Whether alternative EEXO dosing regimens or injection intervals would yield similar results remains unclear. The present protocol, adapted from our previous study [11], involved the IP administration of two consecutive doses of EEXOs (1×10^9 particles per mouse, every 24 h). This approach significantly mitigated LPS-induced liver injury. Our previous study revealed that the in vivo half-life of EEXOs is approximately 48 h [11], suggesting that a single dose of 1×10^9 particles per mouse could exert therapeutic effects. Further studies are necessary to confirm these possibilities.

Second, although the therapeutic effects of EEXO-delivered let-7i-5p were demonstrated in cell-based models and in male mice, its efficacy in female mice and humans remains unknown. Given the well-documented sex-based differences in immune responses [58,59], the magnitude or durability of let-7i-5p-mediated TLR4 suppression may differ between the two sexes. Estrogen and testosterone modulate TLR4 expression and NF- κ B signaling [60,61], which are central to the mechanistic pathways identified in this study. Emerging evidence further suggests that biological sex can influence exosome pharmacology [62,63]. In mice, intravenously administered exosomes exhibit sex-dependent biodistribution, with males showing greater accumulation in the liver, kidney, and brain, and females showing higher uptake in the lungs [62]. Studies of nanoparticle pharmacokinetics indicate that sex hormones can alter immune clearance, vascular permeability, and tissue uptake [63], providing a mechanistic rationale for potential sex-dependent differences in exosome biodistribution. Moreover, preclinical studies have reported sex-dependent differences in the therapeutic efficacy of exosome-based interventions [64,65], further supporting this concept. As this study used only male mice, sex-related variations in the biodistribution and therapeutic efficacy of EEXO-delivered let-7i-5p cannot be excluded, and future investigations in female animals and sex-stratified human cohorts are warranted.

Third, we focused exclusively on let-7i-5p, without comparing its effects to other microRNAs that confer protection in inflammatory and septic models. For instance, miR-146a and miR-21 suppress NF- κ B

signaling [43,44], and miR-206-3p mitigates LPS-induced acute lung injury by inhibiting the TLR4/nucleotide-binding oligomerization domain leucine-rich repeat and pyrin domain containing 3 axis [45]. A head-to-head comparison between let-7i-5p and other microRNAs would clarify their relative efficacy and therapeutic potential for inflammation-induced organ injury.

Finally, although TLR4 emerged as a direct target of let-7i-5p, we did not comprehensively explore other pathways regulated by this microRNA. Members of the let-7 family modulate a wide range of immune mediators, and a study revealed their involvement in the following pathways: mitogen-activated protein kinase/activating protein-1, TIR-domain-containing adapter-inducing interferon- β /interferon regulatory factor 3, and cytokine receptor signaling cascades [66]. Moreover, beyond TLR4, bioinformatic analyses predict that let-7i-5p may also regulate multiple downstream effectors. For instance, miRPathDB and miRTarBase identify high mobility group box 1 (HMGB1), signal transducer and activator of transcription 3 (STAT3), suppressor of cytokine signaling 1 (SOCS1), and interleukin-1 receptor-associated kinase 1 (IRAK1) as potential targets of let-7i-5p (data not shown). These molecules play pivotal roles in inflammation, fibrosis, and immune regulation [67–70], and dysregulation of their signaling pathways has been implicated in sepsis, acute and chronic lung injury, liver fibrosis, and cancer-related inflammation [67–74]. As this study did not evaluate the effects of let-7i-5p on these pathways, potential parallel or synergistic mechanisms cannot be excluded. Future studies integrating unbiased transcriptomic and proteomic profiling, followed by pathway-specific validation, are warranted to delineate these interactions and strengthen the mechanistic understanding of let-7i-5p-mediated therapeutic effects.

5. Conclusion

We generated and rigorously characterized EEXOs, which retained canonical exosomal features and had an approximately 11-fold higher let-7i-5p concentration than did BEXOs. EEXOs were efficiently internalized by RAW264.7 cells, increasing the intracellular concentration of let-7i-5p by approximately 3-fold without affecting viability. Mechanistically, let-7i-5p directly targeted the 3'UTR of TLR4. In LPS-stimulated macrophages, EEXOs suppressed TLR4 upregulation, attenuated MyD88/IRAK1/NF- κ B activation, and reduced TNF- α and IL-6 release. In vivo, systemic administration of EEXOs mitigated LPS-induced acute liver injury in male mice, as reflected by an improvement in histology (lower Suzuki's scores) and reductions in AST and ALT levels, tissue edema, and TLR4 expression at both mRNA and protein levels. Moreover, the inhibition of let-7i-5p abrogated these protective effects, supporting an on-target mechanism.

Our findings should be interpreted with caution because the present study was limited to short observation periods (24 h in vitro and 48 h in vivo), male mice, and a single dosing regimen. Mechanistic analyses focused only on the MyD88/NF- κ B axis, without assessing other relevant pathways. Furthermore, no direct comparisons were made with other microRNAs. Future studies should optimize dosing and delivery; evaluate pharmacokinetics, biodistribution, durability, and safety; and extend investigations to female animals and additional sepsis and organ injury models. Broader mechanistic analyses and comparative or combinatorial testing of microRNAs, supported by scalable manufacturing and rigorous quality control, are necessary to advance the EEXO platform toward robust preclinical validation and eventual clinical translation for TLR4-associated inflammatory diseases.

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CRedit authorship contribution statement

Hong-Phuc Nguyen Vo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation,

Formal analysis, Data curation, Conceptualization. **Chao-Yuan Chang:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Van Long Le:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Hung-Jen Shih:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Bing-Hua Su:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Van Kim Long Vu:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Chun-Jen Huang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (version 4.0) in the writing process in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Patent Pending: The intervention under investigation in this study involves engineered exosomes enriched with let-7i-5p. This technology is currently under evaluation under American Patent No. 63/557,089, held by Chao-Yuan Chang and Chun-Jen Huang. Aside from this, the authors declare that neither they nor any immediate family members have any financial or personal interest in any organization or entity with a direct stake in the subject matter or materials discussed in this article.

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Data availability

All data from the present study are reported within this article. Bioinformatic predictions of potential mRNA targets of microRNA let-7i-5p, generated using four algorithms—DIANA-microT 2023, miRPATH-DB, miRTarBase, and MiRanda—have been deposited in the Harvard Dataverse (<https://doi.org/10.7910/DVN/IVARXJ>) and are available upon request.

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